
Fault diagnosis
in industrial system
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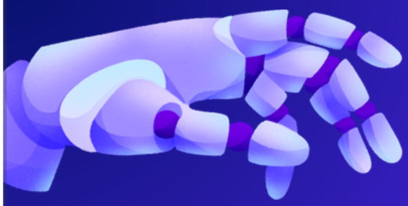
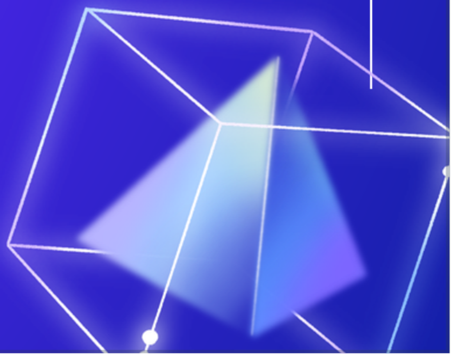


Introduction

Fault diagnosis in industrial systems is the process of detecting and identifying faults in machines and industrial equipment. It helps improve safety, reliability, and performance by finding problems early and reducing machine failures. Fault diagnosis is widely used in manufacturing, power plants, robotics, and automation industries.



Abstract

- Fault diagnosis is used to detect and identify faults in industrial systems.
- It helps improve safety, reliability, and efficiency of machines.
- Sensors and monitoring systems are used to detect abnormal conditions.
- It reduces machine downtime and maintenance costs.
- Fault diagnosis is widely used in manufacturing, power plants, robotics, and automation industries.

PEAS Framework for Fault Diagnosis in Industrial Systems

PEAS FRAMEWORK FOR FAULT DIAGNOSIS IN INDUSTRIAL SYSTEMS

PEAS Element	Description	Fault Diagnosis Example	Implementation / Sensors
Performance Measure	How well the system detects faults and achieves its goals.	• Detect faults accurately • Reduce machine downtime • Improve system reliability • Prevent unexpected failures • Reduce maintenance cost	• Fault detection accuracy (%) • Downtime reduction (hours) • Mean Time Between Failures (MTBF) • Maintenance cost savings • System availability (%)
Environment	The industrial area where the system operates.	• Manufacturing machines • Motors, pumps, turbines • Conveyor belts, gearboxes • Power plants, process industries • Industrial automation systems	• Industrial equipment • Production lines • Operating conditions (temperature, load, speed) • IoT / Industrial environment
Actuators	The actions taken by the system after fault detection.	• Send alerts to operators • Stop or isolate faulty machine • Trigger alarms / notifications • Schedule maintenance • Adjust operating parameters	• Alarm systems • Automatic shutdown systems • Control panels / PLC • Maintenance management system • Actuator / Control signals
Sensors	The inputs used by the system to monitor the industrial equipment.	• Detect vibration, temperature, pressure, current, voltage • Detect noise, speed, torque • Identify abnormal signals or patterns	• Vibration sensors • Temperature sensors • Pressure sensors • Current / Voltage sensors • Acoustic sensors, Speed sensors, IoT sensors

TYPES OF ENVIRONMENTS IN FAULT DIAGNOSIS IN INDUSTRIAL SYSTEMS

Environment Type	Description	Agents	Example in Industrial System
Fully Observable	All system parameters and fault states are completely known through sensors and monitoring systems.	Single / Multi	• All parameters (vibration, temperature, pressure, current, speed) are measured accurately. • Example: Temperature of motor measured exactly by temperature sensor.
Partially Observable	Some system states or faults cannot be directly measured or are affected by noise.	Single / Multi	• Some internal faults (bearing wear, crack, insulation damage) cannot be measured directly. • Only their effects (vibration, noise) are observed.
Deterministic	System behavior is predictable; same cause always gives the same effect.	Single	• Overheating always occurs when motor is overloaded. • If the belt is loose, vibration pattern is predictable.
Stochastic	System behavior is uncertain; same cause can give different results due to random factors.	Single / Multi	• Vibration signals vary due to random noise or varying load conditions. • Sensor readings fluctuate due to environment.
Static	Environment remains unchanged while diagnosis is performed.	Single	• Machine is stopped or running under constant conditions during diagnosis. • Parameters remain constant during testing.
Dynamic	Environment changes while diagnosis is in progress.	Single / Multi	• Machine operating conditions (load, speed, temperature) change with time. • System is running and parameters vary.
Discrete	System states and measurements are in discrete (step-wise) values at specific intervals.	Single / Multi	• Sensor readings are sampled at specific intervals (e.g., every 1 second). • Machine states: ON/OFF, Fault/Normal.
Continuous	System states and measurements vary continuously over time.	Single / Multi	• Vibration, temperature, current, and pressure signals vary continuously. • Monitored in real-time without interruption.

Applications

- Manufacturing machine monitoring
- Power plant equipment fault detection
- Automotive engine diagnosis
- Robotics and automation systems
- Oil and gas pipeline monitoring
- Smart factory and IoT systems

Conclusion

Fault diagnosis in industrial systems is important for improving safety, reliability, and efficiency in industries. It helps detect faults early, reduces machine downtime, lowers maintenance costs, and improves productivity. With the use of sensors, monitoring systems, and AI techniques, industries can achieve smooth and continuous operation.


THANK YOU!